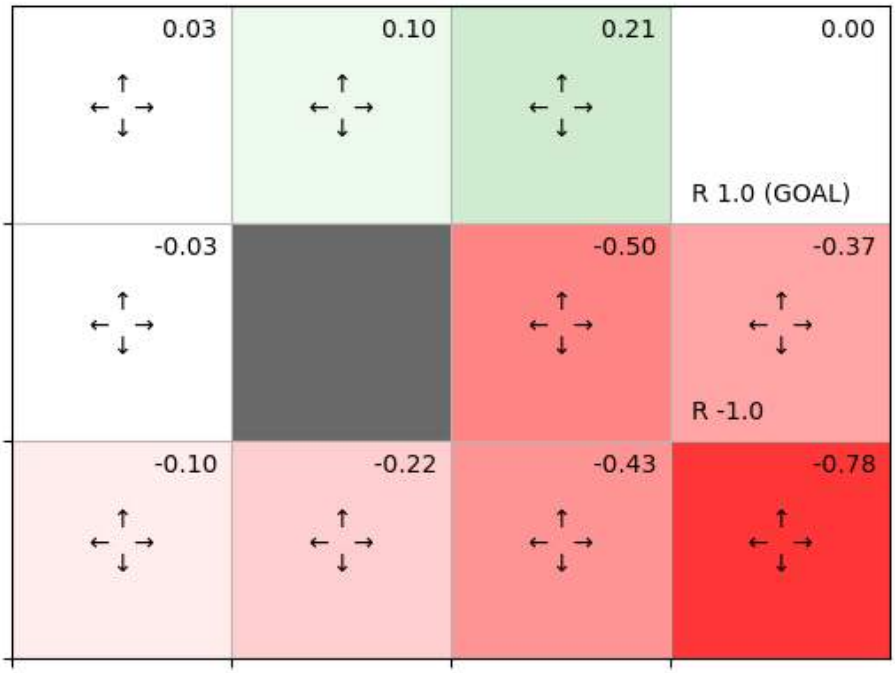
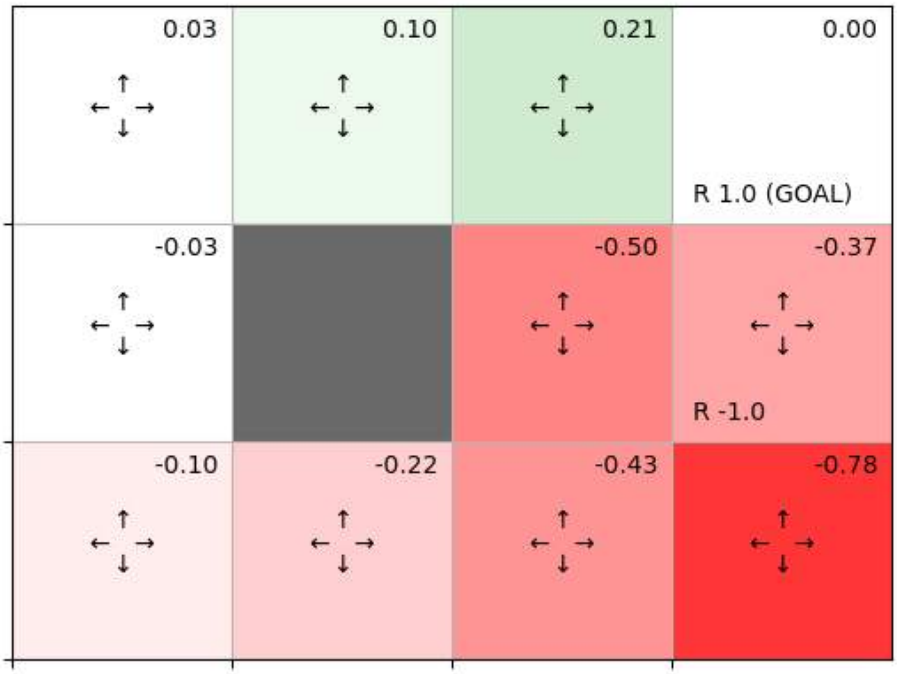
실습 1

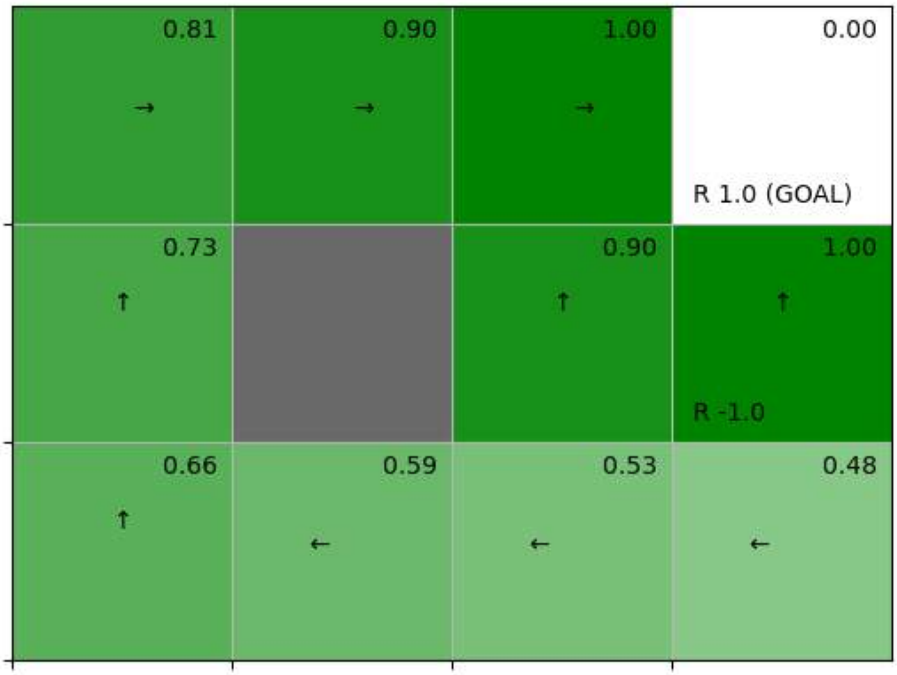
```
from collections import defaultdict
from common.gridworld import GridWorld
def eval_onestep(pi, V, env, gamma=0.9):
    for state in env.states():
        if state ==env.goal_state:
            V[state] =0
            continue
        action_probs =pi[state]
        new_V =0
        for action, action_prob in action_probs.items():
            next_state =env.next_state(state, action)
            r =env.reward(state, action, next_state)
            new_V +=action_prob *(r +gamma *V[next_state])
        V[state] =new_V
    return V
def policy_eval(pi, V, env, gamma, threshold=0.001):
    while True:
        old_V =V.copy()
        V =eval_onestep(pi, V, env, gamma)
        delta =0
        for state in V.keys():
            t =abs(V[state] -old_V[state])
            if t >delta:
                delta =t
        if delta <threshold:
            break
    return V


if __name__=="__main__":
    env =GridWorld()
    gamma =0.9
    pi =defaultdict(lambda: {0: 0.25, 1: 0.25, 2: 0.25, 3: 0.25})
    V =defaultdict(lambda: 0)
    V =policy_eval(pi, V, env, gamma)
    env.render_v(V, pi)
```

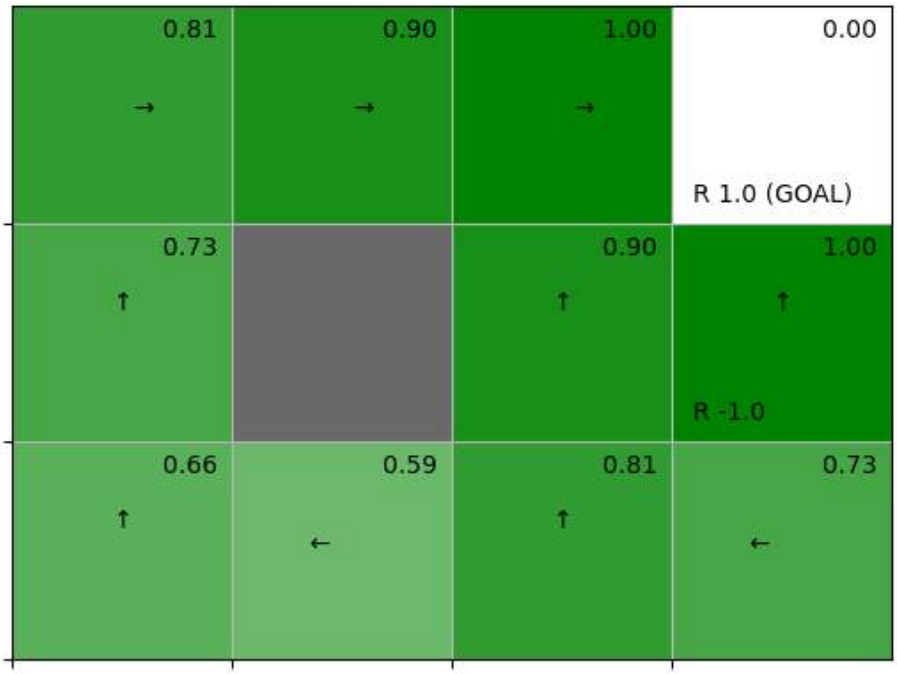

0.03
0.10
0.21
0.00
R 1.0 (GOAL)
-0.03
-0.50
-0.37
R -1.0
-0.10
-0.22
-0.43
-0.78

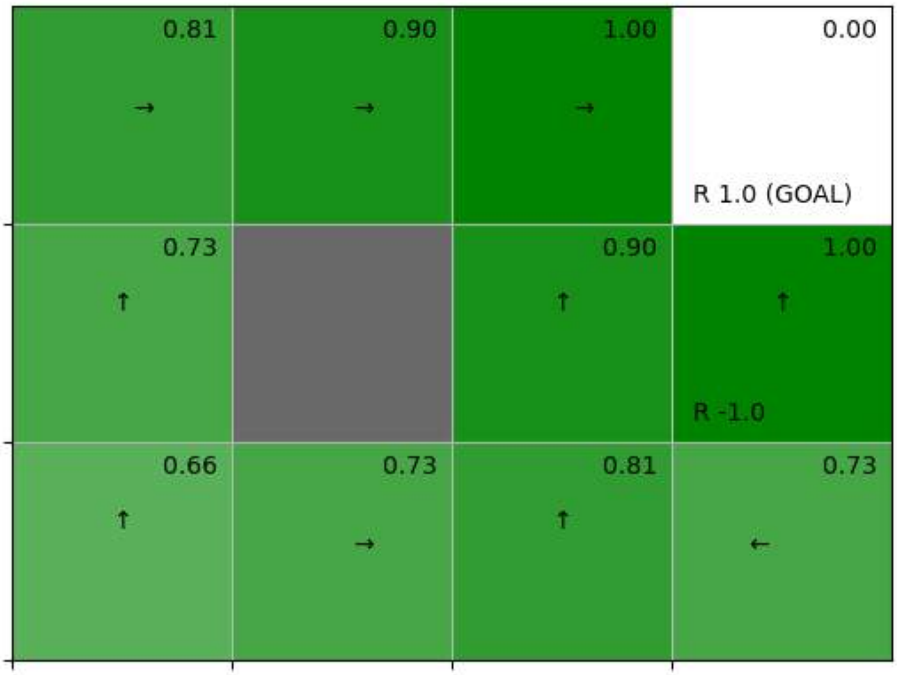
실습 2

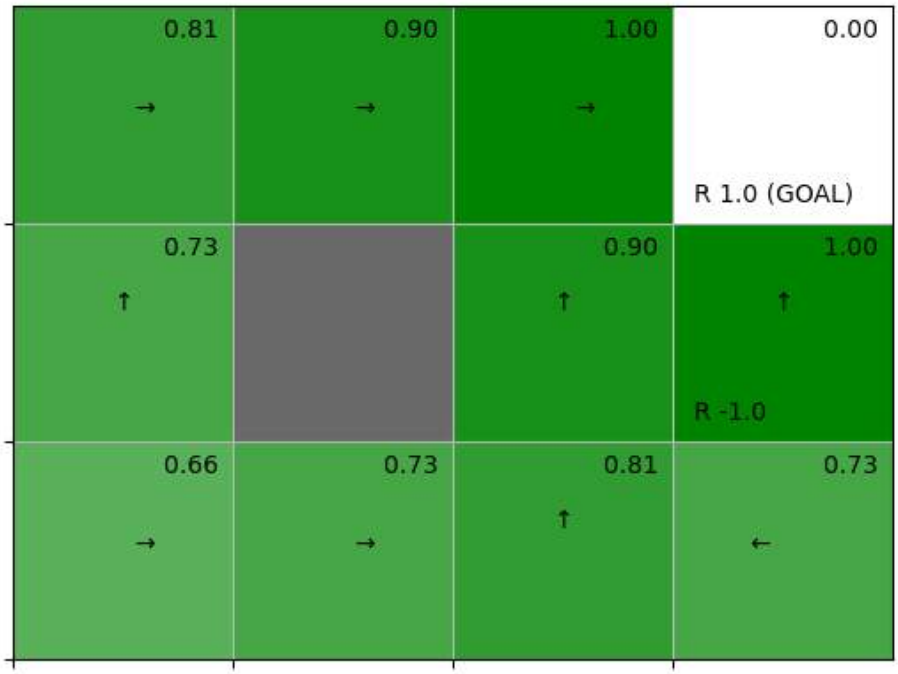
```
from collections import defaultdict
from common.gridworld import GridWorld
from policy_eval import policy_eval
def argmax(d):
    max_value =max(d.values())
    max_key =-1
    for key, value in d.items():
        if value ==max_value:
            max_key =key
    return max_key
def greedy_policy(V, env, gamma):
    pi ={}
    for state in env.states():
        action_values ={}
        for action in env.actions():
            next_state =env.next_state(state, action)
            r =env.reward(state, action, next_state)
            value =r +gamma *V[next_state]
            action_values[action] =value
        max_action =argmax(action_values)
        action_probs ={0: 0, 1: 0, 2: 0, 3: 0}
        action_probs[max_action] =1.0
        pi[state] =action_probs
    return pi
def policy_iter(env, gamma, threshold=0.001, is_render=True):
    pi =defaultdict(lambda: {0: 0.25, 1: 0.25, 2: 0.25, 3: 0.25})
    V =defaultdict(lambda: 0)
    while True:
        V =policy_eval(pi, V, env, gamma, threshold)
        new_pi =greedy_policy(V, env, gamma)
        if is_render:
            env.render_v(V, pi)
        if new_pi ==pi:
            break
        pi =new_pi
    return pi
if __name__=="__main__":
    env =GridWorld()
    gamma =0.9
    pi =policy_iter(env, gamma)
```


0.03
0.10
0.21
0.00
R 1.0 (GOAL)
-0.03
-0.50
-0.37
R -1.0
-0.10
-0.22
-0.43
-0.78


0.81
0.90
1.00
0.00
R 1.0 (GOAL)
0.73
0.90
1.00
R -1.0
0.66
0.59
0.53
0.48

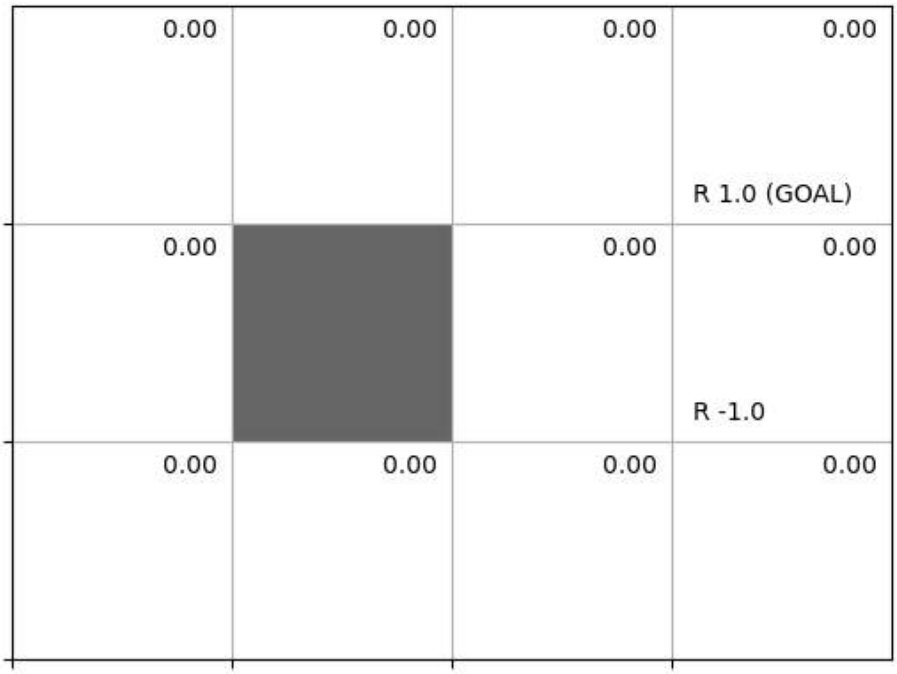

0.81
0.90
1.00
0.00
R 1.0 (GOAL)
0.73
0.90
1.00
R -1.0
0.66
0.59
0.81
0.73

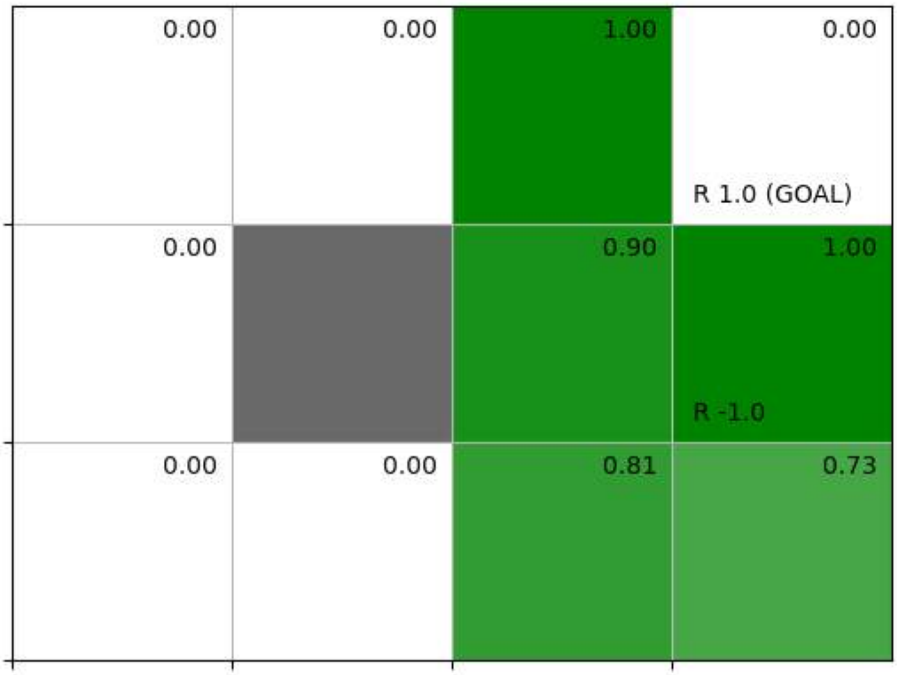

0.81
0.90
1.00
0.00
R 1.0 (GOAL)
0.73
0.90
1.00
R -1.0
0.66
0.73
0.81
0.73

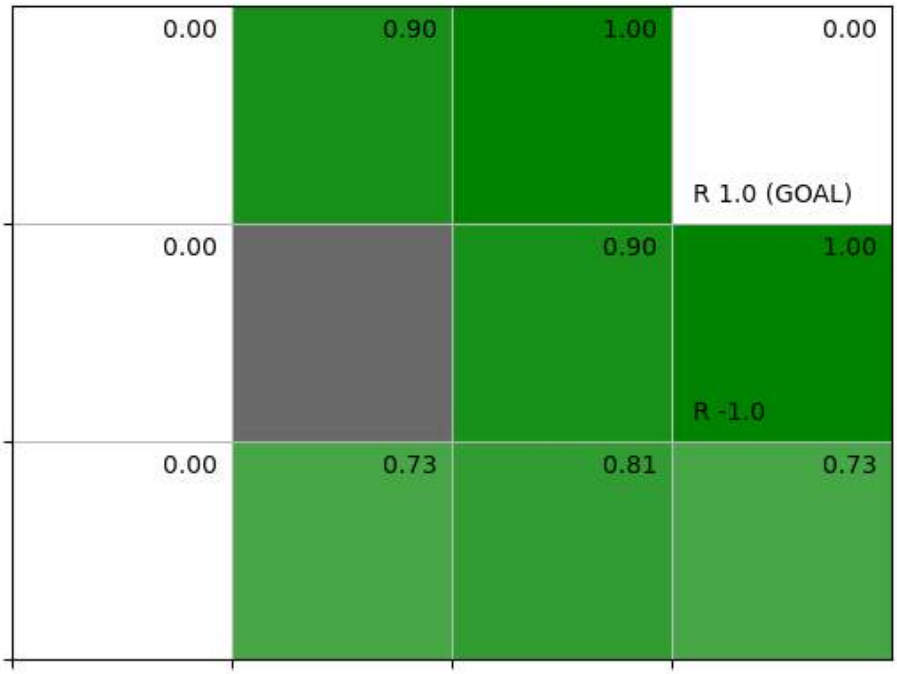

0.81
0.90
1.00
0.00
R 1.0 (GOAL)
0.73
0.90
1.00
R -1.0
0.66
0.73
0.81
0.73

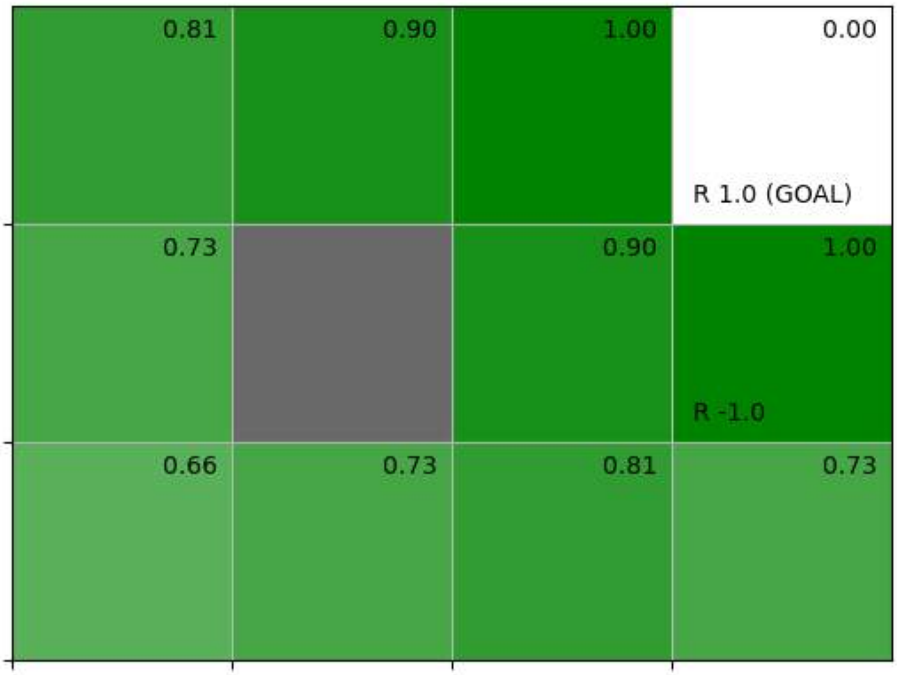
실습 3

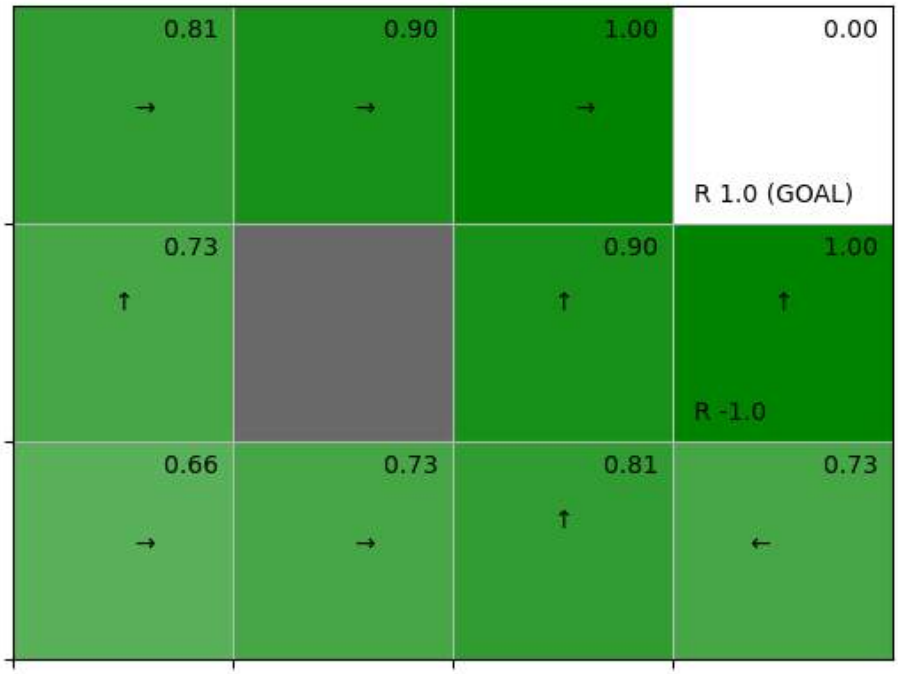
```
from collections import defaultdict
from common.gridworld import GridWorld
from policy_iter import greedy_policy
def value_iter_onestep(V, env, gamma):
    for state in env.states():
        if state ==env.goal_state:
            V[state] =0
            continue
        action_values =[]
        for action in env.actions():
            next_state =env.next_state(state, action)
            r =env.reward(state, action, next_state)
            value =r +gamma *V[next_state]
            action_values.append(value)
        V[state] =max(action_values)
    return V
def value_iter(V, env, gamma, threshold=0.001, is_render=True):
    while True:
        if is_render:
            env.render_v(V)
        old_V =V.copy()
        V =value_iter_onestep(V, env, gamma)
        delta =0
        for state in V.keys():
            t =abs(V[state] -old_V[state])
            if t >delta:
                delta =t
        if delta <threshold:
            break
    return V
if __name__=="__main__":
    V =defaultdict(lambda: 0)
    env =GridWorld()
    gamma =0.9
    V =value_iter(V, env, gamma)
    pi =greedy_policy(V, env, gamma)
    env.render_v(V, pi)
```


0.00
0.00
0.00
0.00
R 1.0 (GOAL)
0.00
0.00
0.00
R -1.0
0.00
0.00
0.00
0.00


0.00
0.00
1.00
0.00
R 1.0 (GOAL)
0.00
0.90
1.00
R -1.0
0.00
0.00
0.81
0.73


0.00
0.90
1.00
0.00
R 1.0 (GOAL)
0.00
0.90
1.00
R -1.0
0.00
0.73
0.81
0.73


0.81
0.90
1.00
0.00
R 1.0 (GOAL)
0.73
0.90
1.00
R -1.0
0.66
0.73
0.81
0.73


0.81
0.90
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0.00
R 1.0 (GOAL)
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R -1.0
0.66
0.73
0.81
0.73